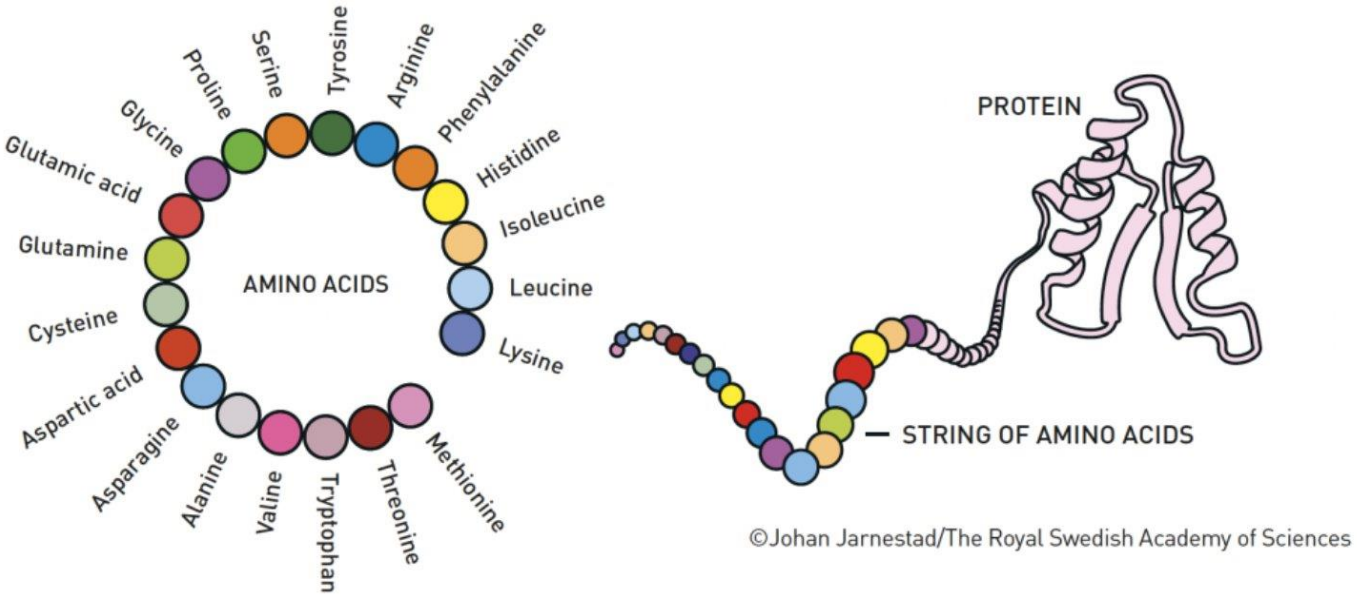
Can I trust AlphaFold?

An analysis guide for predictions

Christine Schulz
Bio-Structure Hub
SSC, IWR

Structure Prediction with AI Models

The AI Revolution with AlphaFold 3



https://www.nobelprize.org/prizes/chemistry/2024/summary/

Nobel Prize in Chemistry 2024



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David Baker

Prize share: 1/2



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Demis Hassabis

Prize share: 1/4



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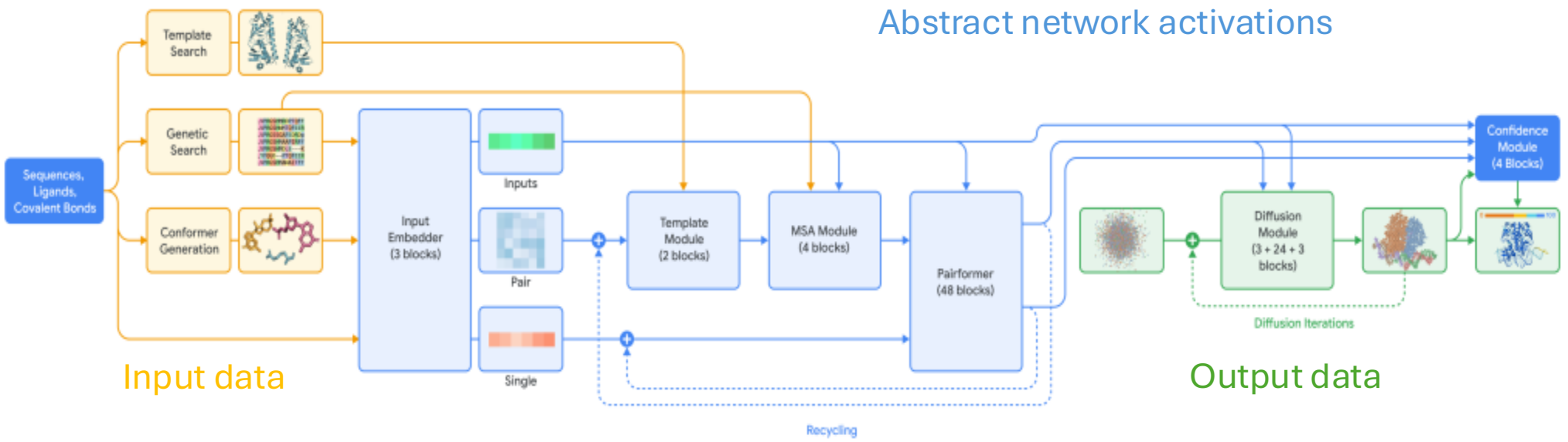
John Jumper

Prize share: 1/4

What is AlphaFold?

Quick Recap

- machine learning model trained on structures in [protein databank]

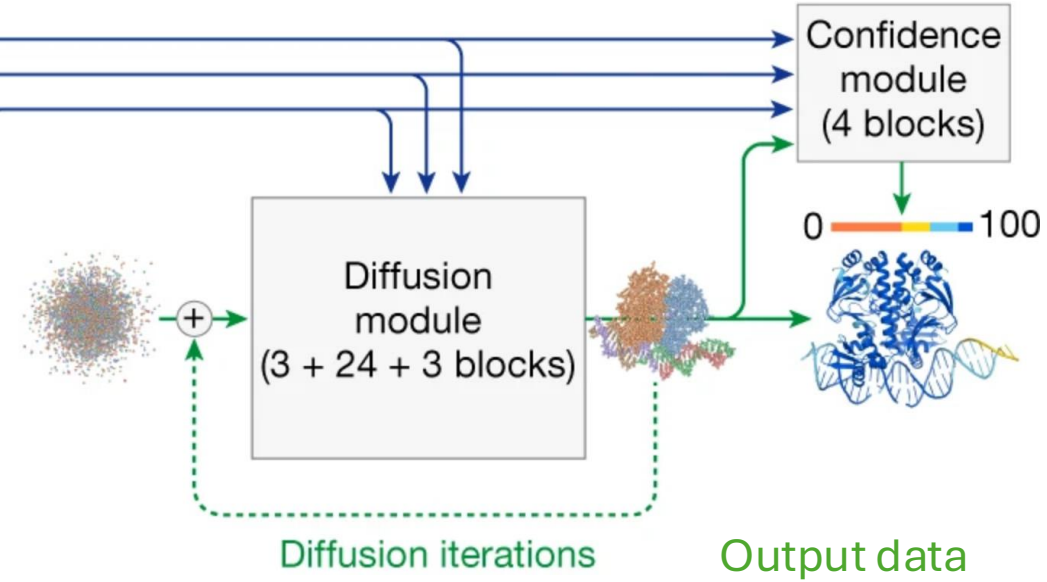


https://www.nature.com/articles/s41586-024-07487-w
https://www.ebi.ac.uk/training/online/courses/alphafold/alphafold-3-and-alphafold-server/introducing-alphafold-3/how-does-alphafold-3-work/

What is AlphaFold?

Quick Recap

- machine learning model trained on structures in [protein databank]



Confidence metrics
- pLDDT
- pTM / ipTM
- PAE

Other metrics / methods
- Swissmodel
- Molecular dynamics

https://www.nature.com/articles/s41586-024-07487-w

What is AlphaFold?

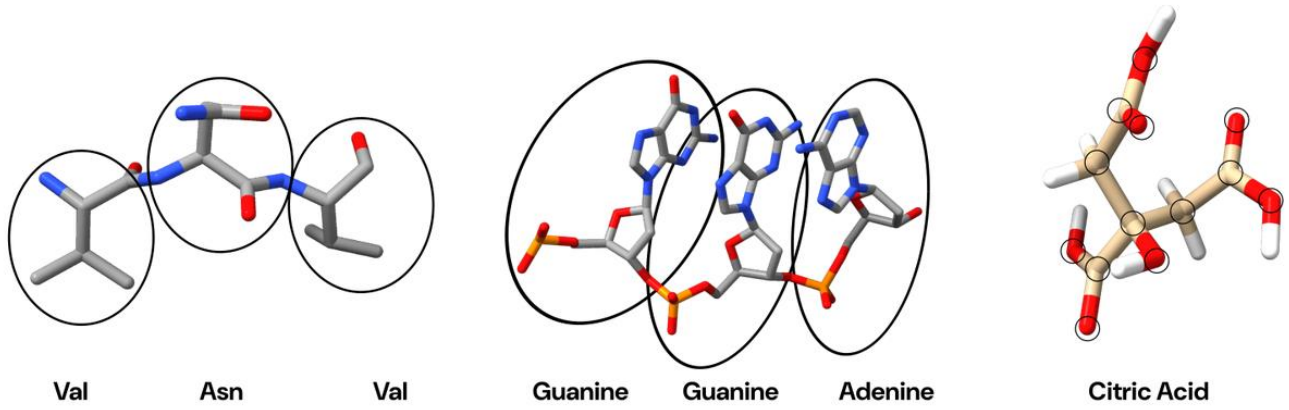
Quick Recap

Tokenization

AF2: Tokens correspond to amino acids

AF3: Tokens depend on the molecule type

- Amino acids, nucleic acids, atoms in small molecules



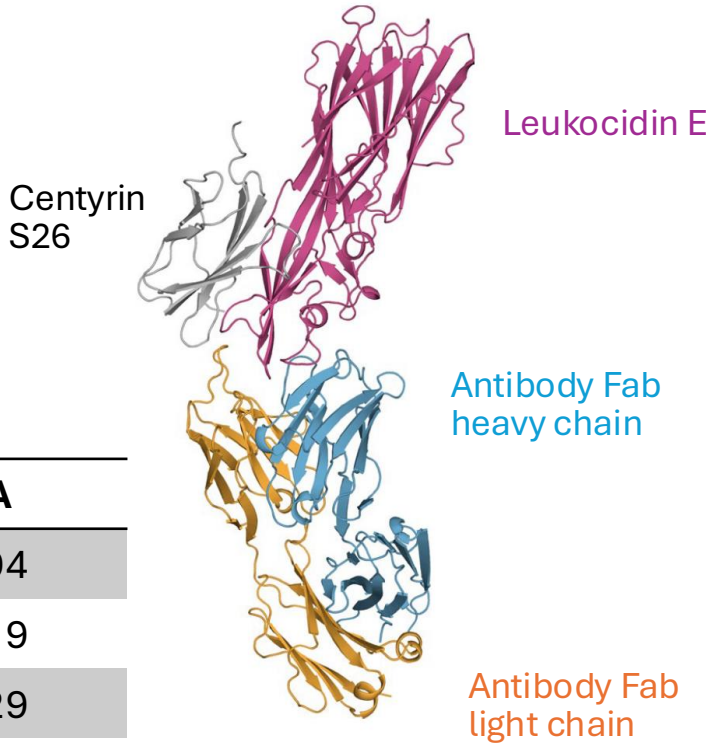
https://www.ebi.ac.uk/training/online/courses/alphafold/

How good is an AlphaFold prediction?

The model system

- PDB reference 7T82
- 4 protein chains
- Antibody light and heavy chain
- Peptide < 100 AA

Chain	AA
Leukocidin E	294
Antibody Fab Light Chain	219
Antibody Fab Heavy Chain	229
Centyrin S26	98



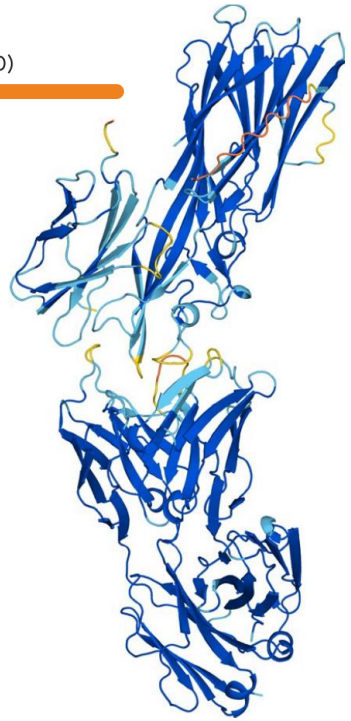
https://doi.org/10.1038/s41586-024-07487-w
https://doi.org/10.1016/j.chom.2023.04.004

How good is an AlphaFold prediction?

plDDT - predicted local distance difference test

Very high (plDDT > 90) | Confident (90 > plDDT > 70) | Low (70 > plDDT > 50) | Very low (plDDT < 50)

- Estimate of position errors between individual atoms and its surrounding (15 Å radius)
- Estimates how well the prediction would agree with an experimental structure
- Measures local confidence of AlphaFold in the correctness of the structure
- pLDDT score can vary significantly along a protein chain: low score expected in flexible regions (termini, turns, loops)



7T82
Leukocidin E
+Antibody Fab
+ Centyrin S26

Abramson, J., Adler, J., Dunger, J., … Jumper, J. M. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, *630*(8016), 493–500. https://doi.org/10.1038/s41586-024-07487-w

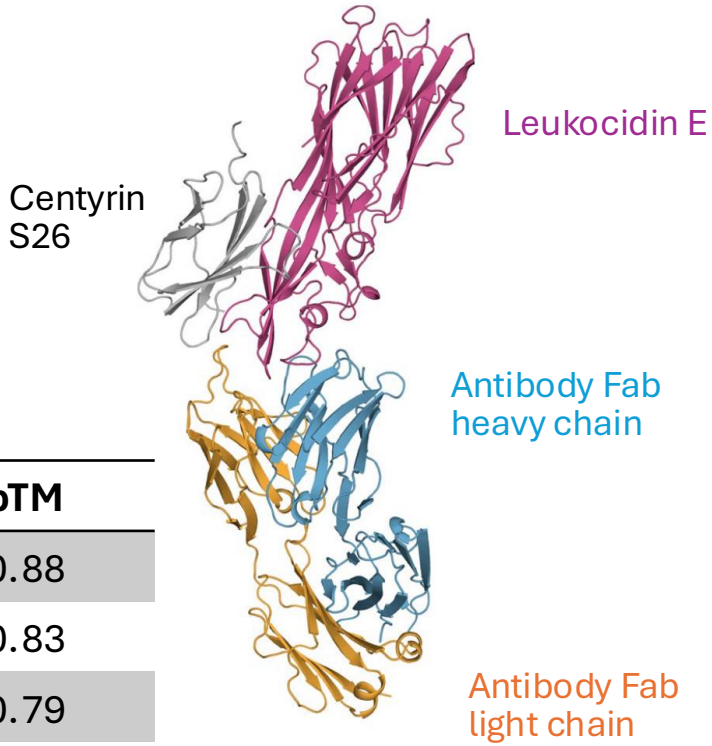
How good is an AlphaFold prediction?

pTM - predicted template modelling

pTM > 0.5 predicted fold similar to the true structure
pTM < 0.5 predicted structure likely wrong

- Accuracy for the whole structure or individual chains
- Less useful for short chains (< 20 tokens)
- Due to averaging, beware of differences in chain lengths between different chains
- disordered regions lead to a decrease in both the pTM and ipTM scores

Chain	pTM
Leukocidin E	0.88
Antibody Fab Light Chain	0.83
Antibody Fab Heavy Chain	0.79
Centyrin S26	0.75
Full structure	**0.54**



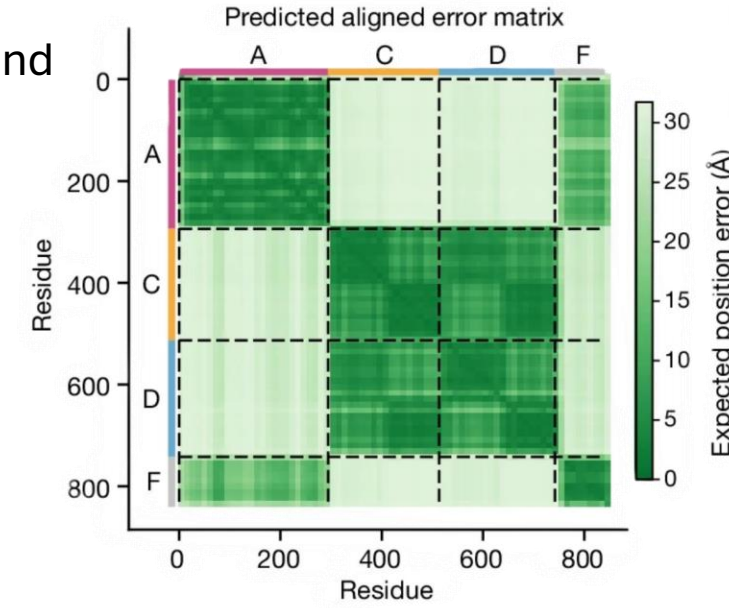
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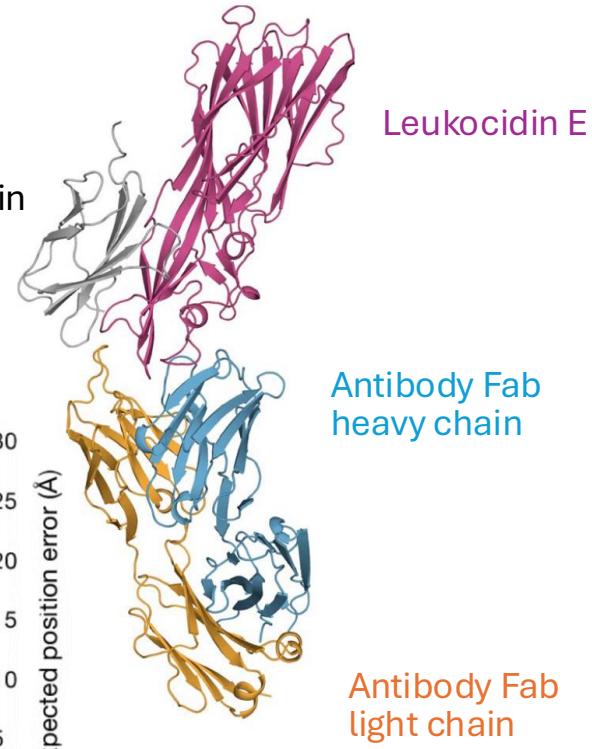
How good is an AlphaFold prediction?

PAE – predicted aligned error

- Measures confidence in relative positions of two items within the predicted structure
- Calculated for pairs of tokens

- High confidence within chains (diagonal)
- High confidence between antibody light and heavy chain
- Domains within antibody chains visible
- High confidence between Leukocidin and Centyrin



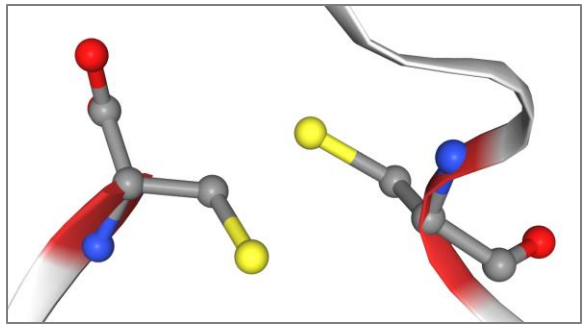


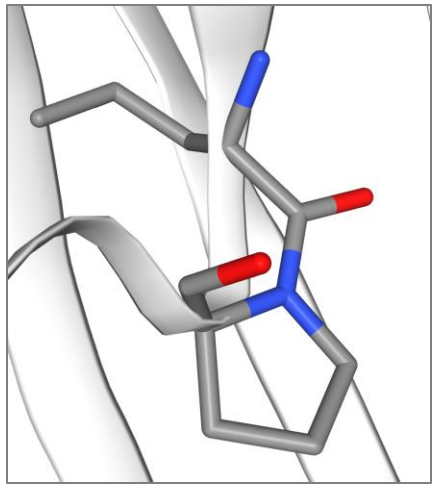
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How good is an AlphaFold prediction?

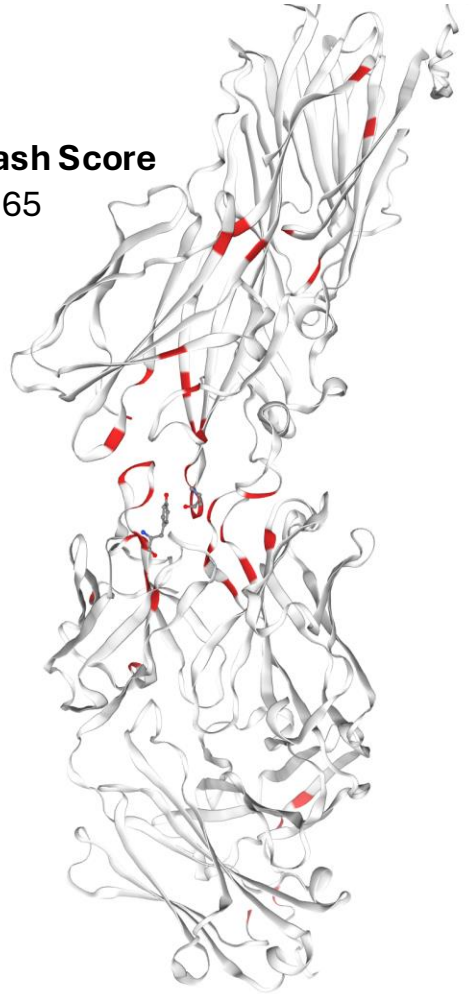
Orthogonal testing

- **Structure Assessment with Expasy assess**
  - Stereochemistry: Clashing bonds / atoms
  - Comparison to reference structures
  - Model quality estimation
  - Ramachandran plot





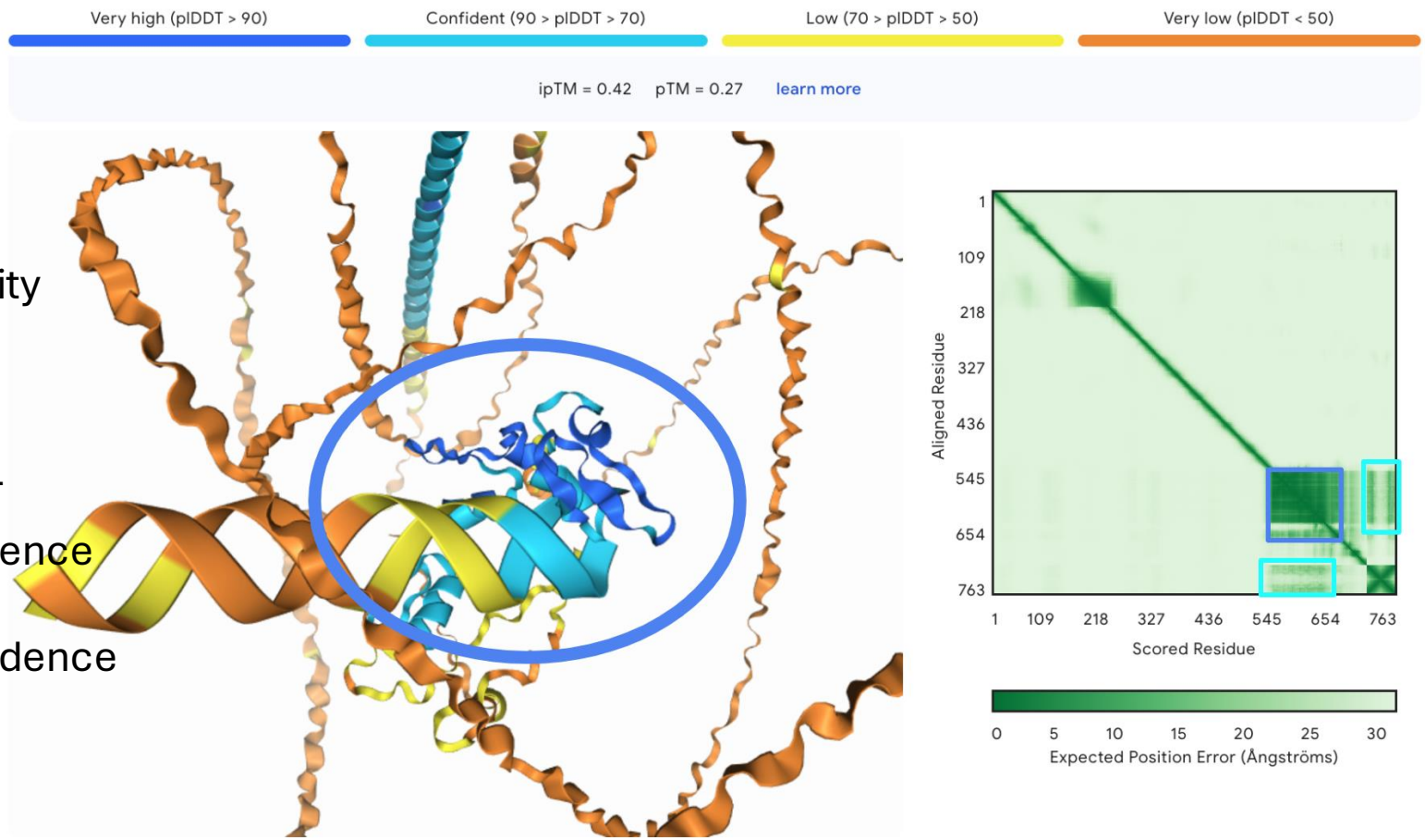
Clash Score
13.65



Waterhouse AM, Studer G, Robin X, Bienert S, Tauriello G, Schwede T (2024) The structure assessment web server: for proteins, complexes and more. *Nucleic Acids Research*, vol. 52, W1 3863480210.1093/nar/gkae270

How to deal with ambiguous confidence?

Combine confidence values



Swi5 protein + DNA

- pTM 0.27
  → low overall quality
- ipTM 0.42
  → low quality of interactions
- Largely low plDDT
  → low local confidence
- High PAE
  → low global confidence

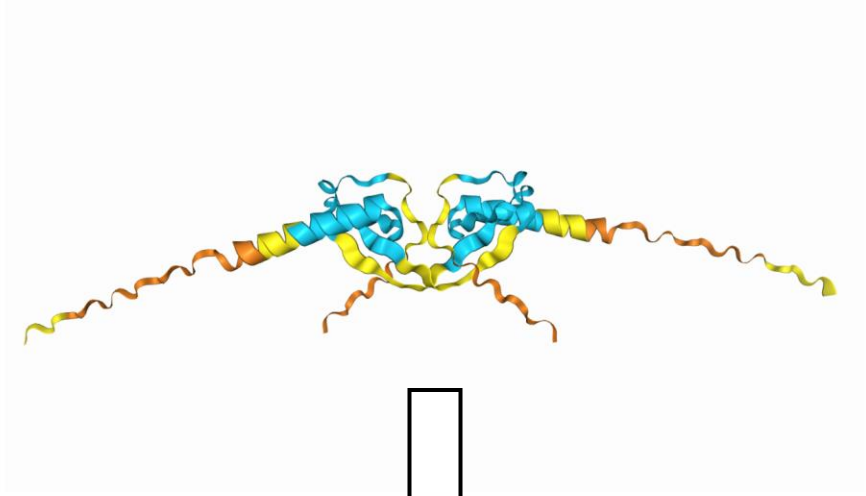
… but:

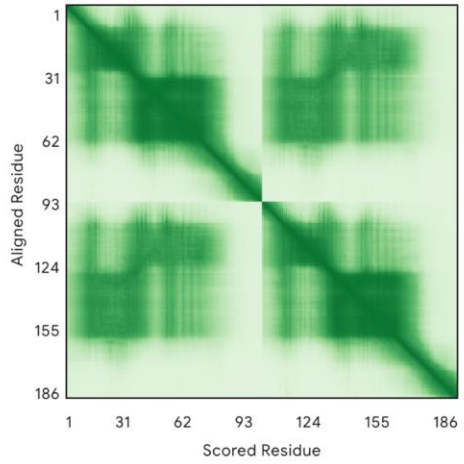
- Very high plddt in binding domain in Swi5
- High pLDDT in corresponding DNA region
- Low PAE for corresponding residues

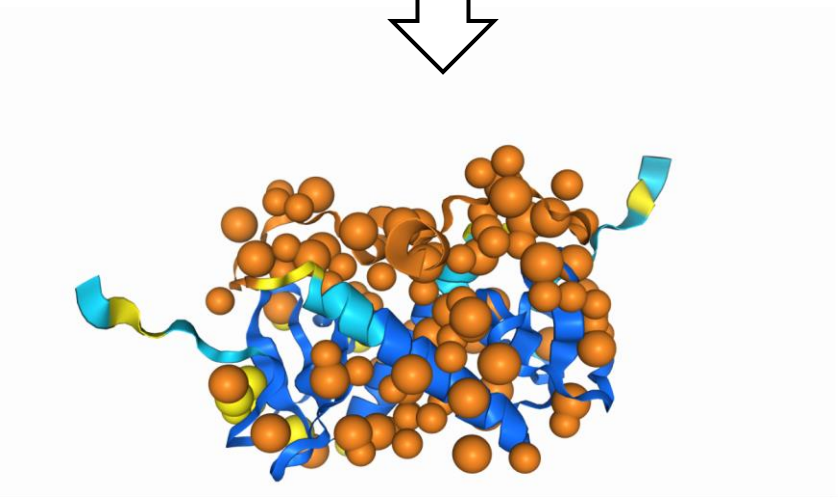
https://alphafoldserver.com/guides#section-3:-interpreting-results-from-alphafold-server

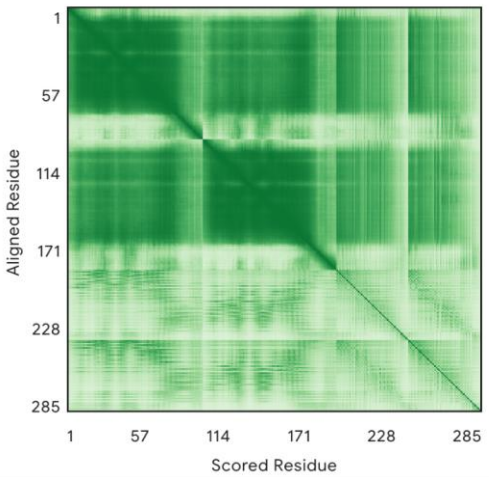
How to improve structure predictions?

Add salts, ligands, cofactors





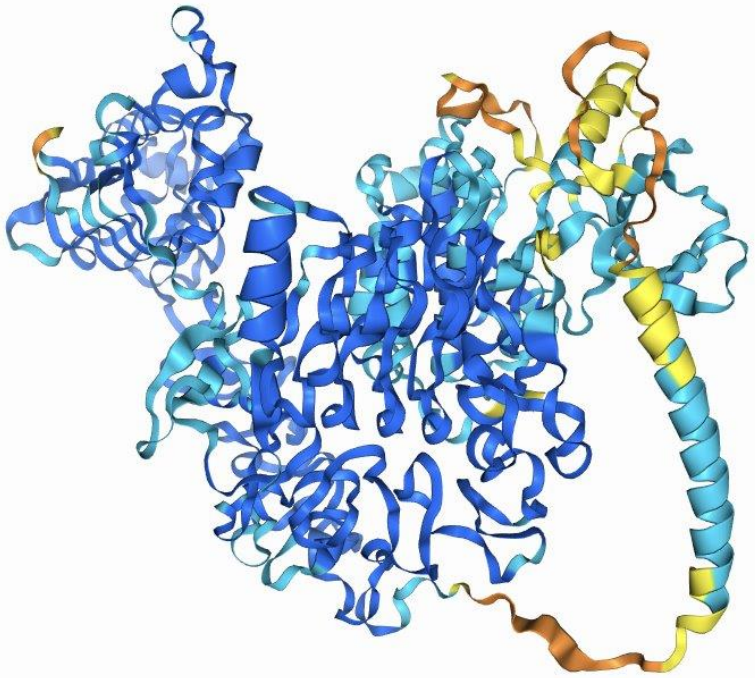


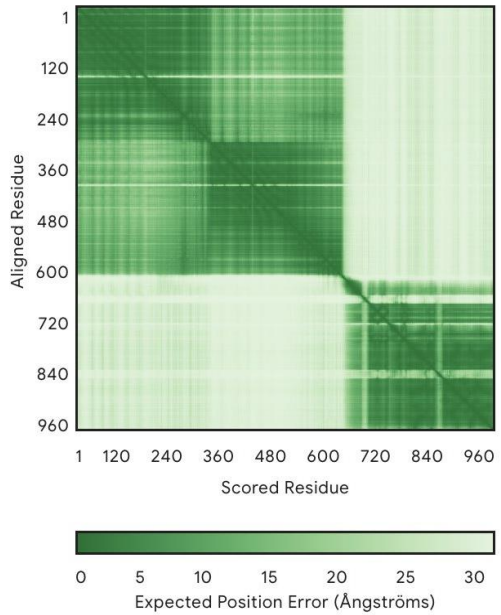


Some structures benefit greatly from added environment

How to improve structure predictions?

Add salts, ligands, cofactors





Add some lipids!

